ENGINEERING DESIGN AND MODELLING

NAME: V.TARUN **EX NO: 2.1**

EX NO: 2.1, 3D SOLID MODEL FROM ISOMETRIC DRAWING.

1. AIM

To create a 2D sketch design using Autodesk Fusion 360 by applying sketch tools, circles, arcs, lines, trimming, mirroring, dimensions, and constraints according to the given engineering drawing.

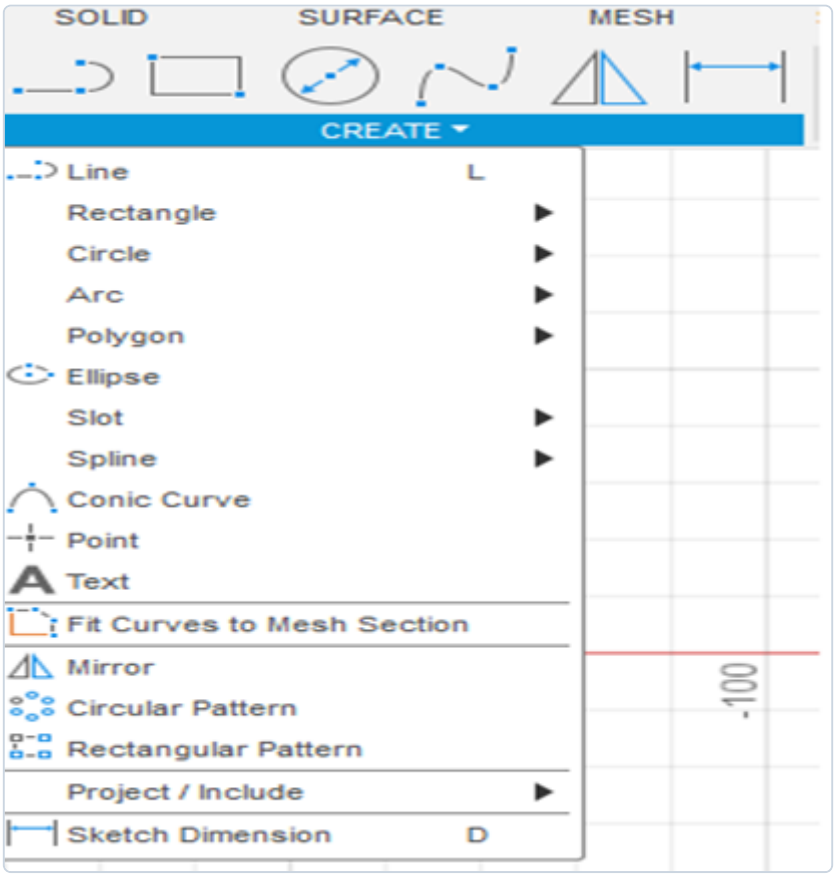


Figure: Autodesk Fusion 360 Create Sketch & Toolbar Interface

2. SOFTWARE AND TOOLS REQUIRED

1. **Software:** Autodesk Fusion 360
2. **Tools required:**

   - Create Sketch
   - Circle Tool
   - Line Tool

- Arc Tool
- Sketch Dimension
- Trim Tool
- Mirror Tool
- Constraint Tools

3. PROCEDURE

1. Start the sketch command:
Select `Sketch > Create Sketch`.

2. Select the Sketch Plane:
Select the Front (XZ) Plane.

3. Create a Rectangle:
Select `Sketch > Rectangle > Center Rectangle`. Hover over the origin (center) of the sketch. Click once to begin placing the rectangle. Press the **Tab** button and enter the values 100 mm and 60 mm for the width and height respectively. Click again to place the rectangle. Click Finish Sketch after completing.

4. Extrude Rectangle in XZ Plane:
Select Extrude. Select the Rectangle sketch. Type in the value of 30 mm. Press ENTER to accept the value.

5. Start the sketch command:
Select `Sketch > Create Sketch`.

6. Select the Sketch Plane:
Select the front surface of the rectangular block.

7. Create a CIRCLE:
Select `Sketch > Circle > Center Circle`. Hover over the origin and click to place the circle. Type the Tab button and enter the value DIA 50 mm. Click again to place the circle. Click Finish Sketch after completing.

8. Extrude CIRCLE:
Select Extrude. Select the circle sketch. Type in the value of 20 mm. Press ENTER to accept the value.

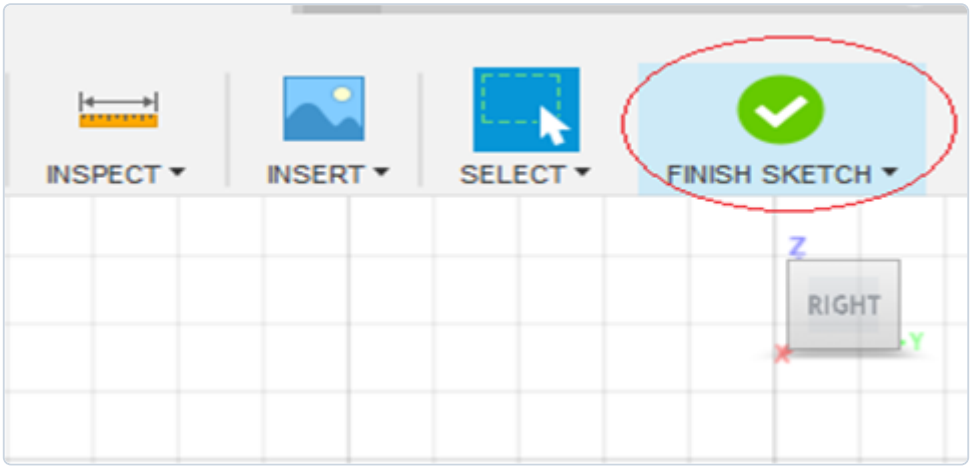


Figure: Finish Sketch Command Button

9. Start the sketch command:

Select `Sketch > Create Sketch`.

10. Select the Sketch Plane:

Select the top surface of the rectangle.

11. Create lines & projections:

Select the Line command. Draw the projections as per the diagram. Click Finish Sketch and Extrude the sketch.

4. OUTPUT

1. Sketch & 3D Model View:

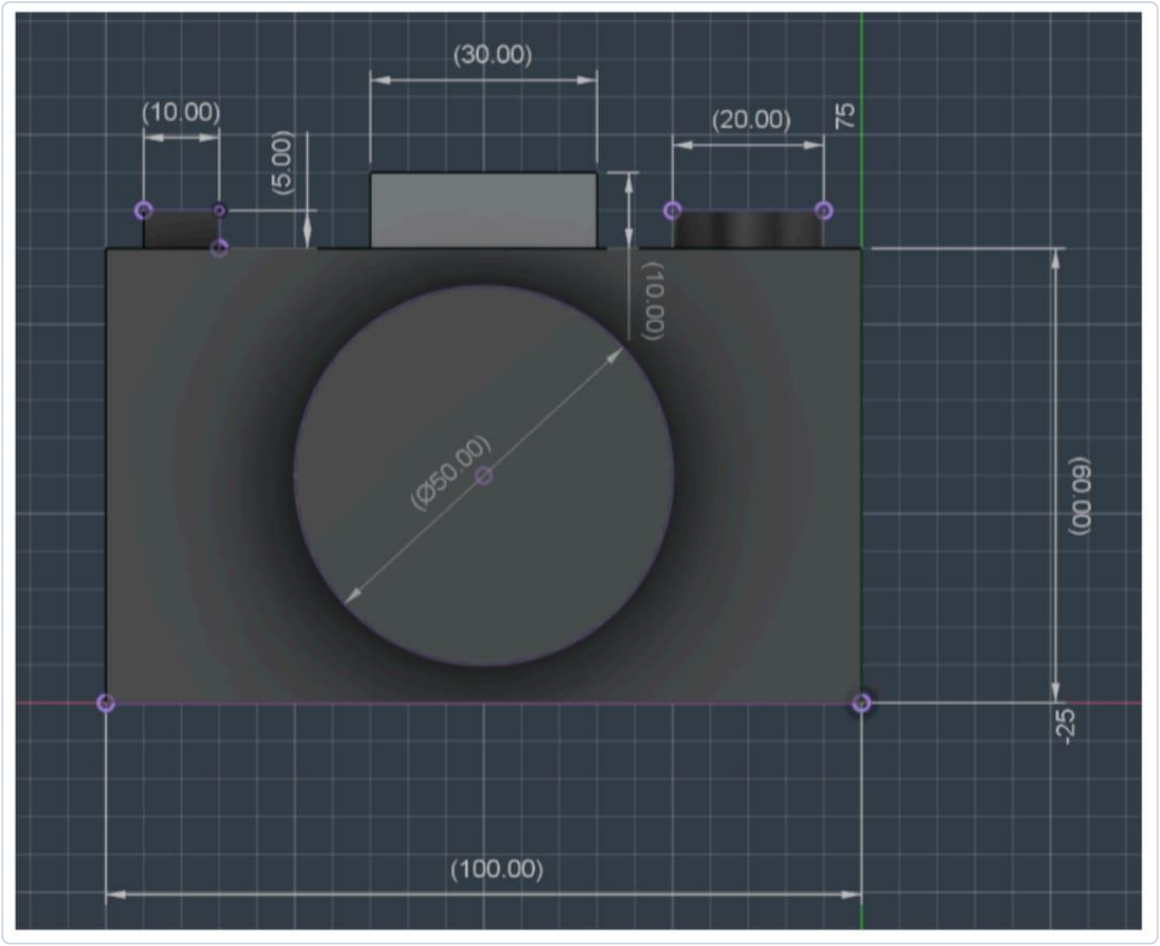


Figure: 3D Model Sketch with Dimensions

2. Design Sketch & Drafting Sheet (with updated Name V.TARUN):

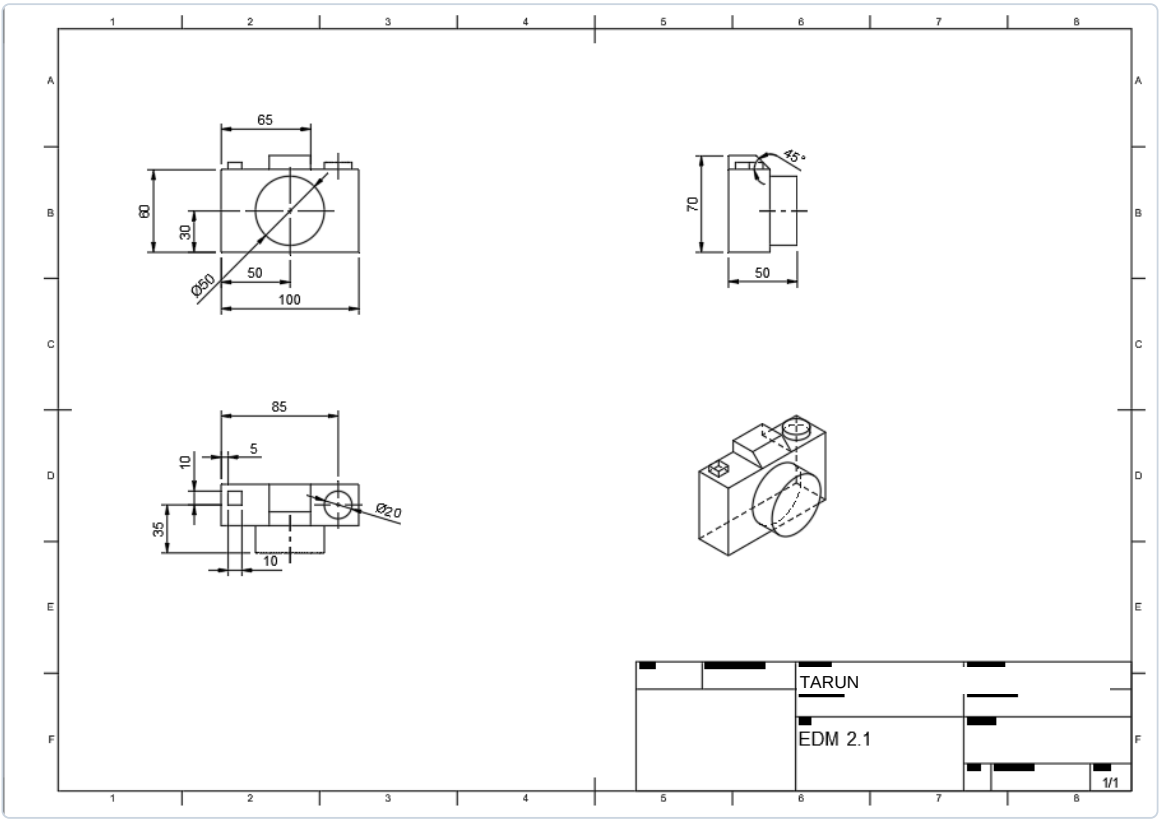


Figure: Final Engineering Drawing Sheet and Orthographic Projections

5. RESULT

Thus, the given model is designed and drafted using Autodesk Fusion 360 using sketch tools, circles, arcs, dimensions, constraints, trimming and mirroring according to the given engineering drawing.

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